

Mathematics For Computing

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Definition of a Set

Definition

A **set** is a well-defined collection of distinct objects called **elements**.

Notation

$$A = \{1, 2, 3\}, \quad 2 \in A, \quad 5 \notin A$$

Example

$$V = \{a, e, i, o, u\}$$

(Set of vowels)

1. Roster Form

$$A = \{2, 4, 6, 8, 10\}$$

2. Set Builder Form

$$A = \{x \in \mathbb{N} \mid x \text{ is even, } x \leq 10\}$$

3. Interval Form

$$[1, 5] = \{x \in \mathbb{R} \mid 1 \leq x \leq 5\}$$

Set Builder Form

A set builder form defines a set by describing the property that every element of the set satisfies.

Notation:

$$\{x : P(x)\} \quad \text{or} \quad \{x \mid P(x)\}$$

where

- x – element of the set
- $:$ or $\mid$ means “such that”
- $P(x)$ – property satisfied by x

Example 1

Natural numbers less than 6

$$A = \{x \in \mathbb{N} : x < 6\}$$

Roster form:

$$A = \{1, 2, 3, 4, 5\}$$

Even numbers up to 10

$$B = \{x \in \mathbb{N} : x \text{ is even and } x \leq 10\}$$

Roster form:

$$B = \{2, 4, 6, 8, 10\}$$

Example 2

Prime Numbers

$$P = \{x \in \mathbb{N} : x \text{ is prime}\}$$

Roster form:

$$P = \{2, 3, 5, 7, 11, \dots\}$$

Multiples of 3

$$M = \{x \in \mathbb{Z} : 3 \mid x\}$$

Roster form:

$$M = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

Example 3

Interval Representation

$$\{x \in \mathbb{R} : 0 < x < 5\} = (0, 5)$$

Solution Set

$$\{x \in \mathbb{R} : x^2 = 9\} = \{-3, 3\}$$

Positive Integers

$$\{x \in \mathbb{Z} : x > 0\} = \{1, 2, 3, \dots\}$$

A **finite set** is a set that contains a limited (fixed) number of distinct elements.

If a set A has n elements for some non-negative integer n , then

$$|A| = n.$$

Examples

- $A = \{1, 2, 3, 4, 5\}, \quad |A| = 5$
- $B = \{a, b, c\}, \quad |B| = 3$
- $C = \{0\}, \quad |C| = 1$
- $\emptyset = \{\}, \quad |\emptyset| = 0$

Definition

An **infinite set** is a set that has infinitely many elements and cannot be counted by any finite number.

A set A is infinite if it is **not finite**.

$$|\mathbb{N}| = \aleph_0$$

where $\aleph_0$ (aleph-null) denotes the cardinality of a countably infinite set.

Examples

- Natural numbers:

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$$

- Even numbers:

$$E = \{2, 4, 6, 8, 10, \dots\}$$

- Integers:

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

- Real numbers:

$$\mathbb{R} = \{x : x \text{ is a real number}\}$$

Subset and Superset

Definition (Subset):

A set A is called a *subset* of a set B , written $A \subseteq B$, if every element of A is also an element of B .

$$A \subseteq B \iff (\forall x)(x \in A \Rightarrow x \in B)$$

Definition (Superset):

A set A is called a *superset* of a set B , written $A \supseteq B$, if every element of B is also an element of A .

$$A \supseteq B \iff B \subseteq A$$

Examples:

Let

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}.$$

Then,

$$A \subseteq B$$

and

$$B \supseteq A.$$

Definition

A set is called a **countable set** if it is finite or if its elements can be put into one-to-one correspondence (bijection) with the set of natural numbers $\mathbb{N}$.

$$A \text{ is countable} \iff |A| \leq \aleph_0$$

Examples

- Finite set:

$$A = \{2, 4, 6, 8\}$$

- Natural numbers:

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

- Integers:

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

- Rational numbers:

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$$

A **singleton set** is a set that contains exactly one element.

$$A = \{a\}, \quad |A| = 1$$

Example

Examples:

- $\{5\}$ is a singleton set.
- $\{a\}$ is a singleton set.
- $\{\{1, 2\}\}$ is also a singleton set because it contains only one element, namely the set $\{1, 2\}$.

Definition

A **power set** of a set A , denoted by $\mathcal{P}(A)$, is the set of all subsets of A .

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

If $|A| = n$, then

$$|\mathcal{P}(A)| = 2^n.$$

Examples

Example 1:

Let

$$A = \{1, 2\}.$$

Then,

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Example 2:

Let

$$B = \{a, b, c\}.$$

Then,

$$\mathcal{P}(B) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

Definition

A **universal set**, denoted by U , is the set containing all the elements under consideration in a particular context.

$$A \subseteq U$$

for every set A in that context.

Examples

Example 1:

$$U = \{1, 2, 3, 4, 5\}, \quad A = \{2, 4\}$$

Then,

$$A \subseteq U.$$

Example 2:

$$U = \{a, e, i, o, u\}, \quad B = \{a, e\}$$

Then,

$$B \subseteq U.$$

Equivalent Sets

Definition

Two sets A and B are called **equivalent** (or **equipotent**) if there exists a bijective (one-to-one and onto) function from A to B .

Mathematically,

$$A \sim B \iff \exists f : A \rightarrow B \text{ such that } f \text{ is bijective.}$$

Notation:

$$A \sim B \quad \text{or} \quad |A| = |B|.$$

Definition

Two sets A and B are said to be **equal** if they contain exactly the same elements.

$$A = B \iff (\forall x)(x \in A \iff x \in B)$$

That is, every element of A belongs to B , and every element of B belongs to A .

Examples

Example 1

$$A = \{1, 2, 3\}, \quad B = \{3, 2, 1\}$$

Since both sets contain the same elements,

$$A = B.$$

Example 2

$$A = \{x \in \mathbb{Z} \mid x^2 < 4\}, \quad B = \{-1, 0, 1\}$$

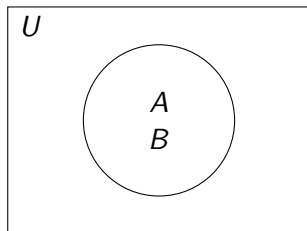
Since

$$A = \{-1, 0, 1\} = B,$$

therefore,

$$A = B.$$

Venn Diagram: Equality of Sets



Since both sets occupy exactly the same region,

$$A = B.$$

Definition

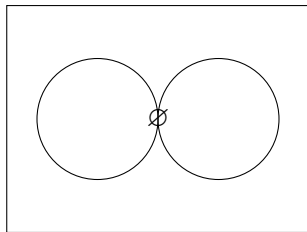
Null / Empty Set

A null (empty) set is a set containing no elements.

Notation:

$$\emptyset, \varnothing, \{\}$$

Venn Diagram



Example

$$\{x \in \mathbb{Z} \mid 1 < x < 2\} = \emptyset$$

$$x^2 + 1 = 0, x \in \mathbb{R} \Rightarrow \emptyset$$

Definition of a Poset

Definition

A **Partially Ordered Set (Poset)** is a pair $(P, \leq)$ where the relation $\leq$ satisfies:

- ① Reflexive: $a \leq a$
- ② Antisymmetric: $a \leq b$ and $b \leq a \Rightarrow a = b$
- ③ Transitive: $a \leq b, b \leq c \Rightarrow a \leq c$

Example

$$(\mathbb{Z}, \leq)$$

where the usual order is a partial order.

Definition

A Hasse diagram is a graphical representation of a poset using only **cover relations**.

Rules:

- Smaller elements at bottom.
- Greater elements at top.
- Arrow heads are omitted.
- Transitive edges are not drawn.

Example: Hasse Diagram

Poset:

$$P = \{1, 2, 4, 8\}$$

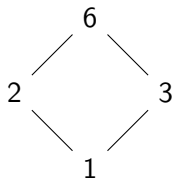


This is a **chain** because every pair of elements is comparable.

Example: Divisibility Poset

For

$$P = \{1, 2, 3, 6\}$$



Cover relations:

$$1 \triangleleft 2, 1 \triangleleft 3, 2 \triangleleft 6, 3 \triangleleft 6$$

Maximum and Minimum Elements

Maximum

$$a \leq m \quad \forall a \in P$$

Minimum

$$n \leq a \quad \forall a \in P$$

Example:

For

$$P = \{1, 2, 3, 6\}$$

Minimum = 1

Maximum = 6

Chains and Antichains

Chain

Every pair of elements is comparable.

Example:

$$\{1, 2, 4, 8\}$$

Antichain

No two elements are comparable.

Example:

$$\{2, 3\}$$

under divisibility.

Inclusion–Exclusion Principle

The Inclusion–Exclusion Principle is used to count the number of elements in the union of sets without counting common elements more than once.

General Form

For n sets $A_1, A_2, \dots, A_n$:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

For Two Sets

For two sets A and B :

$$|A \cup B| = |A| + |B| - |A \cap B|$$

The common elements are subtracted because they are counted twice.

For Three Sets

For three sets A , B , and C :

$$\begin{aligned} |A \cup B \cup C| = & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| \\ & + |A \cap B \cap C| \end{aligned}$$